

Particulate Matter Monitoring & Health Effects

Issue 2026-08-04 | Entry window 4 August 2026 | 13 records in scope

13RECORDS IN SCOPE
(5 off-topic, 5 held)**6**SUBTOPIC CLUSTERS
REPRESENTED**6**RECORDS FROM CHINA
(46% OF CORPUS)**2**EFFECT ESTIMATES
EXTRACTED**2**REVIEWS /
EVIDENCE SYNTHESSES

Signal of the day

Aged Canadian wildfire smoke produced a *negative* Ångström exponent below 500 nm — the sign assumption that underwrites almost every optical PM retrieval broke in the field.

Masoom et al. (*Atmos Chem Phys*) tracked short- and long-range plumes from the 2023 Canadian fire season across North America and Europe with combined remote sensing and in situ measurement. In a roughly week-long episode late in the season they observed **concave spectral curvature in AOD with the maximum above the 340 nm minimum wavelength**, driving the Ångström exponent negative below 500 nm, together with **accumulation-mode volume median diameters reaching ~800 nm** and non-monotonic single-scattering albedo in submicron particles. Each of those three features violates a default that low-cost optical sensing and AOD→PM inversion quietly rely on: that fine smoke is spectrally well-behaved and size-stable. The practical reading for distributed sensing is narrow and concrete — an optical instrument calibrated on background urban aerosol has no valid response function during an aged-smoke episode, and a fixed correction factor will fail in the *direction* of the correction, not merely in magnitude. This is the lesson the regional-calibration literature reached from co-location statistics, arriving here from aerosol physics instead.

What changed today

- ▶ **The health axis is genuinely thin today.** PubMed returned ten health-axis and two sensing-axis records for the 4 August entry date, five of them off-topic. The sensing signal came almost entirely from Crossref-by-journal and Consensus, not PubMed — the structural gap noted since 28 July, unchanged. All six Crossref deposits carry a 4 August created date and every PubMed record an Entrez date of 4 August or earlier, so nothing here belongs to a later issue.
- ▶ **Network siting is emerging as a strand distinct from calibration.** Two independent groups treat sensor placement as an explicit optimisation rather than an afterthought: Magdaong et al. combine street aspect ratio, traffic emissions and irradiance into a suitability raster and reach 23–24% population coverage with 40 nodes in Quezon City; Colarusso et al. define spatial and temporal observability indices for opportunistic vehicle fleets and co-design fixed-node placement against them. Calibration research has matured far faster than siting research, and these are the first two papers this watch has seen that make siting quantitative.
- ▶ **Third consecutive day with no PM-specific ratio estimate carrying a confidence interval.** The two estimates in the forest plot are the O₃ and NO₂ co-pollutant results from Russo et al.; that paper's PM_{2.5} and SO₂ associations were reported only as “less consistent”, without numbers. This is a property of the window, not a pipeline fault — see the figure caption.
- ▶ **Paternal preconception exposure is the more interesting reproductive finding.** Russo et al. window exposure across the 74-day spermatogenesis cycle and resolve it into mitosis, meiosis I–II, spermiogenesis and spermiation — a sharper exposure definition than the trimester blocks this literature normally uses. The signal sits in mitosis and meiosis. Guo et al.'s review independently nominates spermatogenesis stage I (70–90 days pre-collection) as the PM_{2.5} vulnerability window and traffic-derived metal constituents as the driver: convergent, though a review citing the same primary literature is not independent evidence.
- ▶ **Two mechanistic papers, both extra-pulmonary or barrier-focused, both with rescue arms.** Shi et al. place RAGE upstream of PERK/eIF2α/ATF4/CHOP in PM-aggravated COPD and rescue with FPS-ZM1 and 4-PBA; Qi et al. route PM_{2.5} renal fibrosis through alveolar-macrophage IL-6 suppressing ADCY3–CREB–PGC1α and rescue by IL-6 blockade or ADCY3 overexpression. Both use intratracheal instillation, which remains the dosimetric weak point of this whole literature.
- ▶ **Five instrumentation-relevant records were found but not shipped.** Their DOIs verify in Crossref but no abstract is retrievable from Crossref, Europe PMC or Consensus, and the Elsevier and IEEE landing pages are client-rendered. They are logged as held — not rejected — and deliberately kept out of `seen.json` so they can be carried once indexed. Prajapati et al. on PM_{2.5} hygroscopicity in the eastern Indo-Gangetic Plain is the one that matters: RH dependence is directly load-bearing for low-cost sensor correction.

Corpus analytics

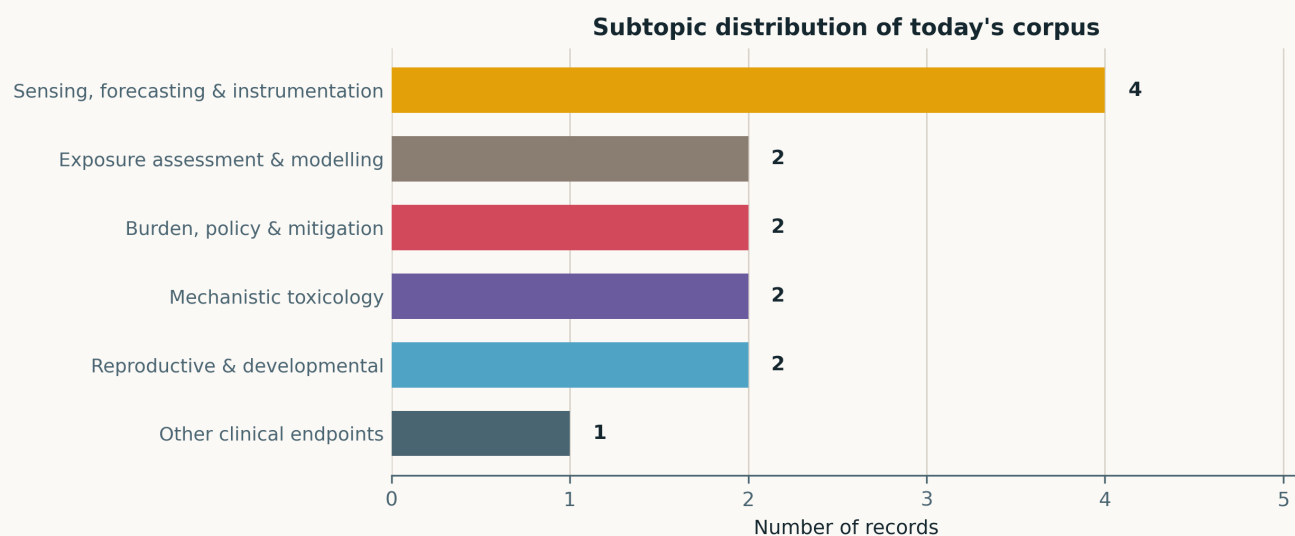


Fig. Subtopic assignment of the 13 in-scope records, single-label by dominant contribution. *Sensing, forecasting & instrumentation* leads with 4 (31%); the remaining five clusters carry one or two records each. Four of the ten standing subtopics — neuro/mental health, cardiovascular/metabolic, respiratory/allergic as a primary label, and occupational/indoor — are empty in this window.

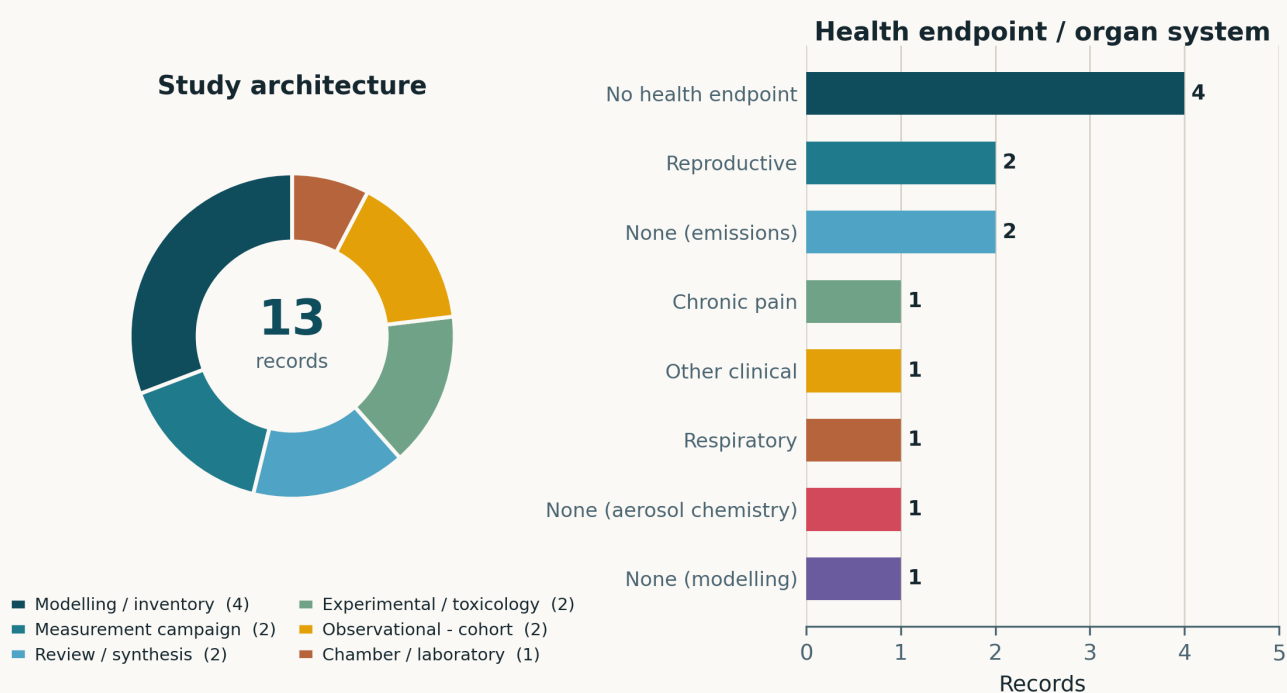


Fig. Left — study architecture. Modelling and inventory work dominates (4/13), ahead of measurement campaigns, reviews, cohort epidemiology and experimental toxicology at two each. **All 13 records initially fell through to "Other / mixed" on this run**; the design map has been extended and an unmapped-design warning added so this failure is loud rather than silent in future (see run log). Right — health endpoint by organ system. Seven of thirteen records carry no human health endpoint at all — four monitoring or modelling, two emissions — the highest non-health share this watch has recorded.

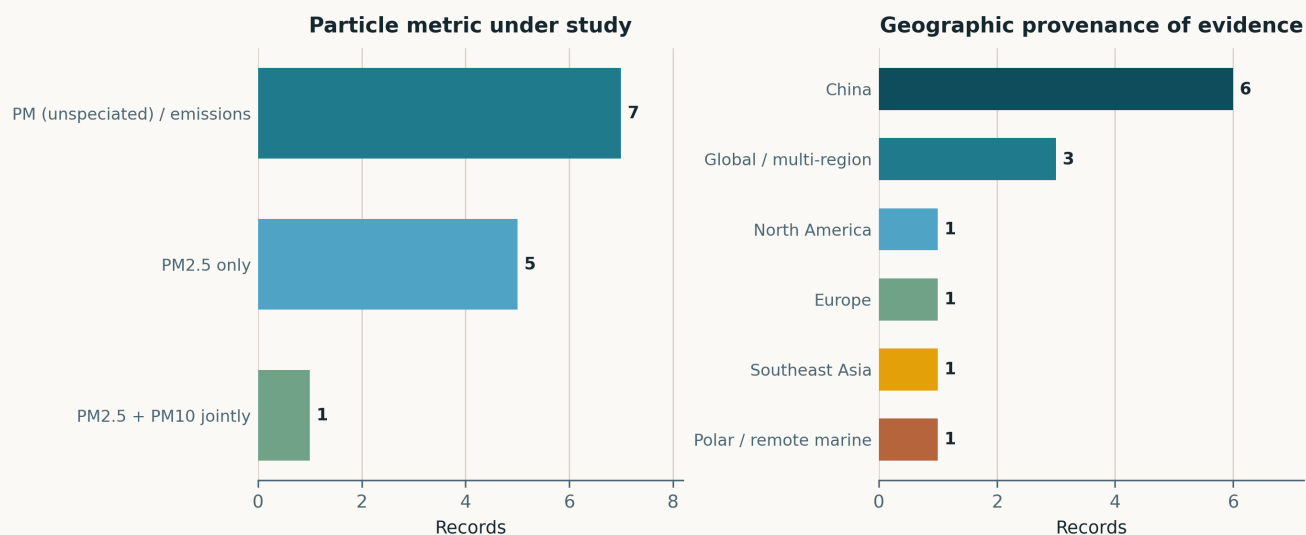


Fig. Left — particle metric. Only five records use PM_{2.5} as the sole exposure and one pairs it with PM₁₀; the remaining seven are unspiciated PM, particle number, INP concentration or column-integrated optical properties. That spread is itself the finding — mass concentration is not the measurand in over half of today's corpus. Right — geographic provenance. China contributes 6 of 13 (46%). Southeast Asia and the Eurasian Arctic each appear once; South Asia and sub-Saharan Africa are absent again.

Life-course windows addressed today (records may span several stages)

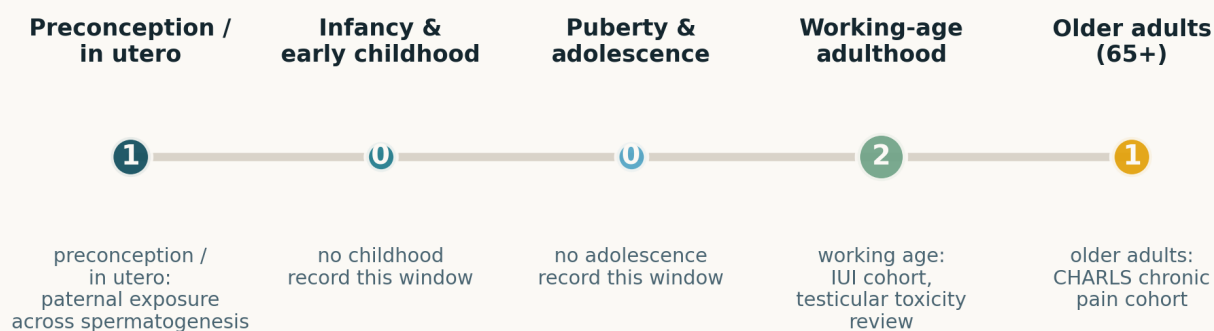


Fig. Life-course windows addressed; counts are non-exclusive, and only four of thirteen records carry a human life-course window at all. The preconception entry is paternal rather than maternal — an unusual and welcome inversion of this literature's default framing.

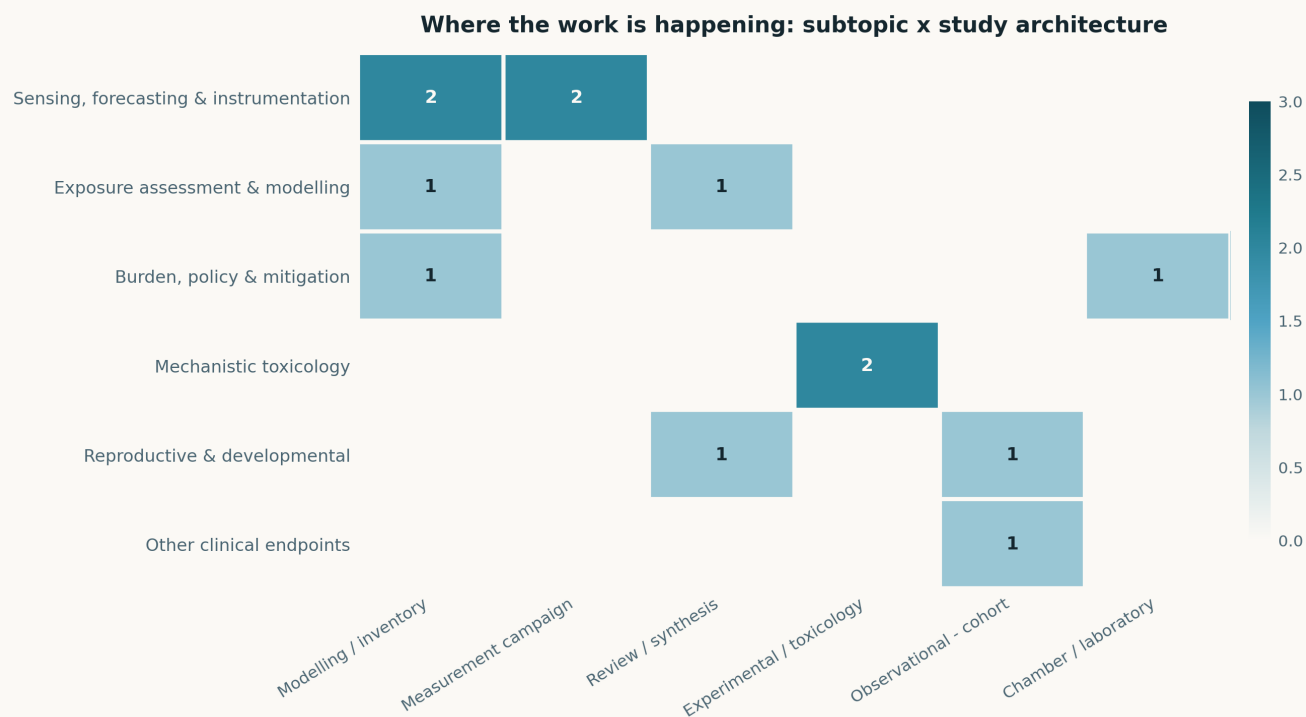


Fig. Subtopic $\times$ study-architecture cross-tabulation. The persistent structural feature is visible again: the *Sensing* row contains no health-endpoint linkage, and the mechanistic row is entirely experimental with no human biomarker bridge. With 13 records spread across six subtopics and six architectures the matrix is necessarily sparse — read it as a coverage map, not a distribution.

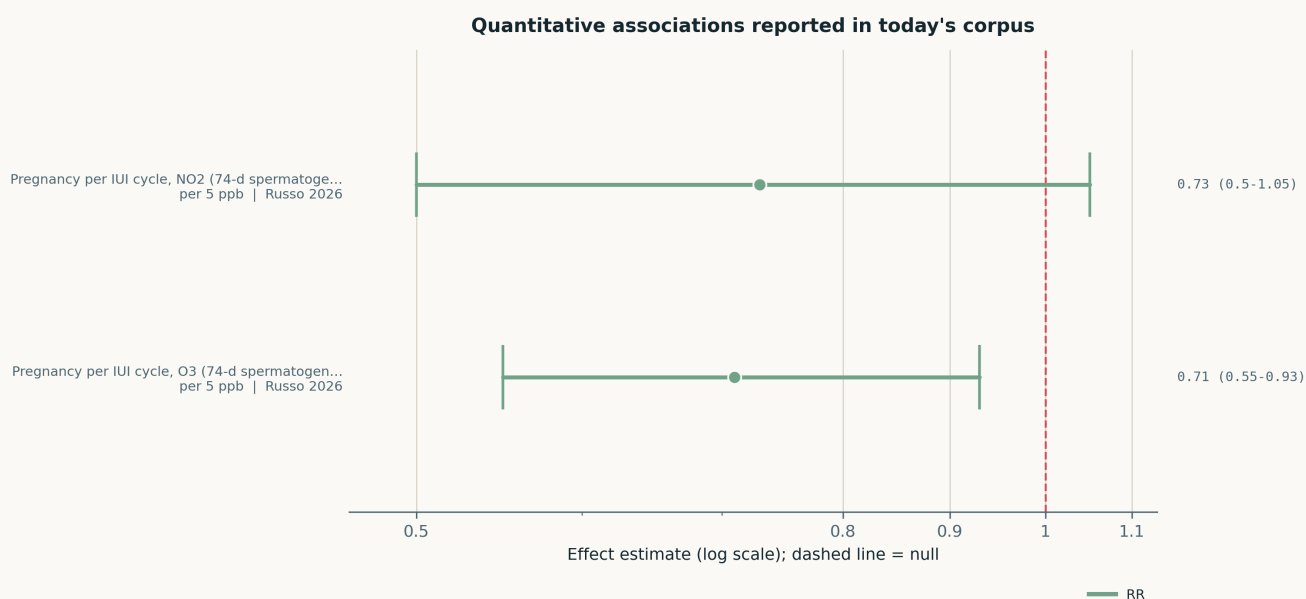


Fig. Both estimates plotted are co-pollutant results, not PM results. No record in this entry window reported a PM_{2.5} or PM₁₀ ratio estimate with a confidence interval — the third consecutive issue for which that holds. Russo et al. report RR 0.71 (95% CI 0.55–0.93) per 5 ppb O₃ and RR 0.73 (0.50–1.05) per 5 ppb NO₂ for pregnancy per IUI cycle across the 74-day spermatogenesis window; their PM_{2.5} and SO₂ associations were characterised as less consistent and given no numeric value. The plot is included so that the absence is recorded explicitly rather than inferred from a missing figure, and so a downstream weekly or monthly rollup does not read three sparse days as data loss.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Sensing, forecasting & instrumentation

4 records

Masoom et al. 2026 Atmos Chem Phys

Long-range transport of Canadian wildfire smoke to Europe in 2023: aerosol properties and spectral features

Combined column remote sensing and in situ measurement of short- and long-range plumes across North America and Europe through the April–October 2023 fire season. The late-season episode is the payload: **concave AOD spectral curvature with the maximum above the 340 nm minimum wavelength, giving negative Ångström exponents below 500 nm**; accumulation-mode volume median diameters near 800 nm; and non-monotonic single-scattering albedo, rising through the UV–visible then flattening or falling at longer wavelengths, consistently across most stations. Limitation: a rare ~one-week event characterised observationally, with no closure between the retrieved optics and a measured composition, so the cause of the curvature is inferred rather than demonstrated. For distributed optical sensing the implication is that smoke-episode response cannot be extrapolated from background calibration.

doi: 10.5194/acp-26-10801-2026

Magdaong et al. 2026 ISPRS Ann Photogramm Remote Sens Spat Inf Sci

Criteria and an algorithm for siting a low-cost IoT air-quality network near roads

GIS multi-criteria analysis combining street aspect ratio, traffic emissions, global horizontal irradiance and road proximity into a 0–6 suitability surface for Quezon City, then a heuristic selecting nodes by joint suitability and population raster under a minimum-spacing constraint. A 40-sensor configuration covers **1.27–1.35 million residents, 23.0–24.4% of the city**, stably across weighting schemes while keeping spatial dispersion balanced. Limitation: coverage is defined geometrically against a population raster with no validation against measured concentration fields, so “covered” means “near a node”, not “well estimated”. Still the clearest statement this watch has seen that siting is an optimisation problem with a stated objective, and the criteria transfer directly to any near-road deployment in a low- or middle-income city.

doi: 10.5194/isprs-annals-x-5-w4-2025-315-2026

Colarusso et al. 2026 Internet of Things

Mobile urban sensing: spatio-temporal observability analysis and optimisation

Fleet-aware framework for hybrid mobile–fixed sensing on an Edge–Fog–Cloud architecture, evaluated on real data from three sensing vehicles covering 11.37 km² in Benevento, Italy. Introduces spatial and temporal observability indices and a value-of-fleet metric, then uses spatio-temporal interpolation plus what-if analysis to quantify how fleet size, route redundancy, sampling density and data loss propagate into reconstructed-map accuracy; a greedy co-design procedure places fixed nodes against the residual. Limitation: a single city, a single small fleet, and interpolation-based ground truth — the observability indices are internally consistent but not externally validated. The framing is the transferable part: opportunistic mobile coverage becomes a measurable quantity rather than a qualitative claim.

doi: 10.1016/j.iot.2026.101907

Li et al. 2026 Atmos Chem Phys

Summertime ice-nucleating particle concentrations over the Eurasian Arctic seas

Shipborne immersion-mode INP measurements from the Arctic Century Expedition across the Barents, Kara and Laptev Seas and adjacent high-Arctic islands, August–September 2021. N_{INP} was lower than at continental high-latitude sites, most clearly below -15°C , implying sparse mineral-dust INP; concentrations peaked in ice-free water, marginal ice zones and near land, and were very low inside the ice pack, with the maximum north of Novaya Zemlya under back-trajectories from the western Siberian coast. The authors conclude sources are local to regional with little long-range transport signature. Limitation: one summer and one cruise track, so the seasonal and interannual envelope is unconstrained. Peripheral to PM exposure, but relevant to interpreting particle counts in clean marine air, where the concentration floor is set by local sources rather than by transport.

doi: 10.5194/acp-26-10835-2026

Exposure assessment & modelling

2 records

Buhendwa et al. 2026 Environ Pollut

Modelling the coupled PM_{2.5}–ozone system in China: mechanisms, model biases, perspectives

Review of the compound-pollution era in which PM_{2.5} and O₃ share a precursor pool (VOCs, NO_x, SO₂, NH₃) and single-pollutant control can worsen the co-pollutant — the “pollution seesaw”. The useful content is the bias inventory: chemical transport models **underestimate secondary organic aerosol by about 40%**, with substantial spread across models, seasons and regions; daytime HONO sources and OH reactivity are missing or understated; ammonia and episodic agricultural-burning inventories are uncertain; nocturnal boundary-layer mixing is poorly represented. These errors propagate into data assimilation and into machine-learning models trained on CTM output. Limitation: a narrative synthesis with no quantitative pooling, so the 40% figure is a reported central tendency, not a meta-analytic estimate. Directly relevant if CTM fields are used as priors or features in a sensor-fusion product.

doi: 10.1016/j.envpol.2026.128887

Zhang et al. 2026 Environ Sci Technol

H₂O₂-driven sulfate formation at air–water interfaces: stepwise mechanism and accelerated kinetics

Theoretical study of the dominant global route to particulate sulfate. At the air–water interface the pre-reaction complex is favoured and the reaction proceeds stepwise — nucleophilic attack of HSO₃[−] on H₂O₂ to a HOOSO₂[−] intermediate, then isomerisation to sulfate — with a **rate-determining barrier of 4.1 kcal/mol**, lower than in bulk, and with the rate-determining step itself shifting. Interfacial electric field and partial solvation together account for **89% of the barrier reduction**. Limitation: entirely computational, with no experimental kinetic validation, and the idealised interface is not a deliquesced particle surface. It matters because rising H₂O₂ from wildfire activity makes this pathway more important, and because sulfate is both a large PM_{2.5} mass fraction and one of its more hygroscopic — therefore optically most humidity-sensitive — components.

doi: 10.1021/acs.est.5c17902

Reproductive & developmental

2 records

Russo et al. 2026 Paediatr Perinat Epidemiol

Critical windows of air-pollution exposure during spermatogenesis and IUI outcomes

Secondary analysis of the Folic Acid and Zinc Supplementation Trial: 592 male partners in the Salt Lake City area contributing **1,223 intrauterine insemination cycles** over nine months. Residential exposure came from Community Multiscale Air Quality models, averaged over the 74-day spermatogenesis window before insemination and separately over mitosis, meiosis I–II, spermiogenesis and spermiation, with generalised linear mixed models adjusted for age, season, income and co-pollutants. Per 5 ppb across the full window, **O₃ RR 0.71 (95% CI 0.55–0.93)** and **NO₂ RR 0.73 (0.50–1.05)** for hCG+ pregnancy, concentrated in mitosis and meiosis; live-birth results were similar and no pattern emerged for pregnancy loss. **PM_{2.5} and SO₂ were less consistently associated and no numeric estimate was given.** Limitation: an infertility-treatment population is not the general population, residential modelling ignores occupational and commuting exposure, and co-pollutant models cannot cleanly separate a traffic mixture. The stage-resolved exposure definition is the methodological contribution worth borrowing.

doi: 10.1111/ppe.70185

Guo et al. 2026 Reprod Toxicol

Environmental pollutants and testicular toxicity: basic evidence and translational prospects

Narrative review of testicular-level evidence for microplastics, bisphenols and air pollution, against a reported >50% decline in global sperm concentration over five decades. The PM-relevant claims: **spermatogenesis stage I, 70–90 days before semen collection, is nominated as the PM_{2.5} vulnerability window**, with metal constituents of traffic-derived PM_{2.5} as the primary toxicity driver. Separately, microplastics are reported in human testicular tissue with a gut-dysbiosis → immune activation → blood–testis barrier pathway, and BCAT1-mediated ferroptosis is offered as a bisphenol injury mechanism. Limitation: no systematic search or quality appraisal, so the window and constituent claims inherit whatever selection bias the cited primary literature carries. Read alongside Russo et al. — convergent on timing, but not independent evidence for it.

doi: 10.1016/j.reprotox.2026.109310

Mechanistic toxicology

2 records

Shi et al. 2026 J Appl Toxicol

PM_{2.5} aggravates airway inflammation and barrier disruption in COPD mice via RAGE-dependent PERK/eIF2α/ATF4/CHOP

Cigarette-smoke-induced murine COPD model with subsequent PM_{2.5} exposure. PM worsened lung function, emphysema, inflammation, apoptosis and epithelial tight-junction damage, and activated the PERK/eIF2α/ATF4/CHOP endoplasmic-reticulum stress arm. The causal claim rests on two rescue arms: the RAGE inhibitor FPS-ZM1 and the ER-stress modulator 4-PBA each attenuated the phenotype, placing RAGE upstream of ER stress rather than parallel to it. Limitation: instilled PM at a dose the abstract does not state, a smoke-primed rather than naive background, and pharmacological inhibitors of imperfect selectivity — FPS-ZM1 in particular. The RAGE→ER-stress ordering is the novel part; the downstream inflammatory readouts are well-trodden.

doi: 10.1002/jat.70358

Qi et al. 2026 J Hazard Mater

Urban PM_{2.5} worsens renal injury and fibrosis through a lung–kidney IL-6/ADCY3 axis

Repeated intratracheal urban PM_{2.5} at an environmentally relevant dose combined with renal ischaemia–reperfusion injury in mice, plus tubular epithelial cell work. PM exposure worsened renal dysfunction, tubular injury, fibrosis and lipid-droplet accumulation; the proposed route is PM → systemic inflammation with alveolar macrophages as the likely IL-6 source → suppression of ADCY3–CREB–PGC1α → impaired fatty-acid β-oxidation, increased glycolysis, mitochondrial dysfunction and fibrosis. IL-6 blockade and ADCY3 overexpression each rescued, in vivo and in vitro. Limitation: the renal-IRI co-insult makes this a second-hit model, so the finding concerns PM aggravating existing injury rather than causing nephropathy; and intratracheal instillation again overstates deposited dose rate. The extra-pulmonary axis and the metabolic mechanism are both worth tracking.

doi: 10.1016/j.jhazmat.2026.143119

Burden, policy & mitigation

2 records

Bu & Sun 2026 J Air Waste Manag Assoc

PM emissions from China's iron and steel industry: patterns, drivers, implications

LMDI decomposition over 99 Chinese sites, separating stack from fugitive emissions into five process factors: steel output, energy intensity, waste-gas generation coefficient, stack concentration and fugitive emission factor. **National PM emissions fell 23.6% from 2011 to 2016.** Stack concentration was the dominant inhibitor (to –31,558 t in 2015–16); the fugitive factor flipped from +5,037 t (2010–11) to –23,320 t; the waste-gas coefficient pushed emissions up throughout (+19,516 t in 2015–16); energy intensity contributed under 25%. Regionally the picture inverts — South and Northwest China show annual increases, with Northwest fugitive emissions reaching 232% of the annual change. Limitation: the series ends in 2016, predating the ultra-low-emission retrofit programme entirely. The stack-versus-fugitive split is what matters for siting monitors near industry: the two have completely different spatial signatures.

doi: 10.1080/10962247.2026.2698603

Zhang et al. 2026 J Hazard Mater

Particle-number emissions from a hybrid electric vehicle across cold and warm starts

A China-6 compliant HEV tested over real-driving-emissions and RTS95 cycles with fresh and bench-aged aftertreatment. The particle-number size distribution was trimodal and dominated by an accumulation mode peaking near **90 nm**, with most emission from urban and highway segments and **engine-start events, cold and warm alike, as the dominant source.** Counter-intuitively, catalyst ageing *improved* overall PN removal: oxidation efficiency fell but gasoline particulate filter filtration rose as an ash layer established. Optimising warm-start fuel enrichment cut PN further on the aged system. Limitation: one vehicle, so this characterises a configuration, not a fleet. Directly relevant to near-road sensing — a 90 nm mode driven by discrete start events is largely invisible to mass-based instruments.

doi: 10.1016/j.jhazmat.2026.143156

Other clinical endpoints

1 record

Yang et al. 2026 J Pain*Long-term exposure to multiple air pollutants and chronic pain in middle-aged and older Chinese adults*

China Health and Retirement Longitudinal Study, 2011–2020: **10,378 participants free of chronic pain at baseline, 8,047 incident cases over a mean 7.39 years** — an incidence approaching 78%, itself the study's most important number and a strong hint that the outcome definition is broad. Time-dependent Cox and accelerated failure time models were combined with weighted quantile sum regression over the five-pollutant mixture. PM_{2.5}, PM₁₀, SO₂ and CO were each associated with higher risk and earlier onset, and **SO₂ was the predominant mixture contributor**, not PM. Limitation: no interval estimates appear in the abstract, self-reported pain invites differential misclassification, and in a highly correlated mixture WQS weights identify the best-measured proxy as readily as the causal agent — read "SO₂ drives it" as a hypothesis.

doi: 10.1016/j.jpain.2026.106400

Corpus register

First author	Focus	Journal	Design	Metric	Region
Reproductive & developmental					
Guo et al.	Environmental pollutants and testicular toxicity in male reproductive health: basic evidence and translational prospects	<i>Reprod Toxicol</i>	Review / synthesis	PM _{2.5} (traffic-derived metal constituents)	Global / multi-region
Russo et al.	Critical windows of air pollution exposure during spermatogenesis and intrauterine insemination outcomes	<i>Paediatr Perinat Epidemiol</i>	Cohort (secondary trial analysis)	PM _{2.5} , NO ₂ , O ₃ , SO ₂ (CMAQ, 74-day spermatogenesis window)	USA
Mechanistic toxicology					
Qi et al.	Exposure to urban PM _{2.5} exacerbates renal injury and fibrosis via the lung-kidney axis through IL-6/ADCY3 signalling	<i>J Hazard Mater</i>	Animal + in vitro	Urban PM _{2.5} (repeated intra-tracheal)	China
Shi et al.	PM _{2.5} exacerbates airway inflammation and epithelial barrier disruption in COPD mice through the RAGE-dependent PERK/eIF2a/ATF4/CHOP pathway	<i>J Appl Toxicol</i>	Animal experiment	PM _{2.5} (in-stilled)	China
Sensing, forecasting & instrumentation					
Colarusso et al.	Mobile urban sensing: spatio-temporal observability analysis and optimization	<i>Internet of Things</i>	Method / algorithm	Reconstructed pollutant concentration maps	Italy
Li et al.	Summertime ice-nucleating particle concentrations over the Eurasian-Arctic Seas	<i>Atmos Chem Phys</i>	Field campaign / shipborne	INP number concentration (immersion mode)	Arctic Ocean
Magdaong et al.	Developing criteria and an algorithm for a low-cost IoT-based air quality sensor network for near-road air quality monitoring	<i>ISPRS Ann Photogramm Remote Sens Spat Inf Sci</i>	Method / algorithm	PM (low-cost sensor network siting)	Philippines
Masoom et al.	Long range transport of Canadian wildfire smoke to Europe in 2023: aerosol properties and spectral features of smoke particles	<i>Atmos Chem Phys</i>	Field campaign / remote sensing	AOD; Angstrom exponent; single scattering albedo; volume size distribution	Multi-country
Exposure assessment & modelling					
Buhendwa et al.	Modeling the coupled PM _{2.5} -ozone system in China: chemical mechanisms, persistent model biases, and future perspectives	<i>Environ Pollut</i>	Review / synthesis	PM _{2.5} and O ₃ (chemical transport models)	China
Zhang et al.	H ₂ O ₂ -driven sulfate formation at air-water interfaces: stepwise mechanism and accelerated kinetics	<i>Environ Sci Technol</i>	Theory / simulation	Particulate sulfate formation rate	Computational
Burden, policy & mitigation					
Bu & Sun	Particulate matter emissions from the iron and steel industry of China: patterns, drivers, and implications	<i>J Air Waste Manag Assoc</i>	Emission inventory / decomposition	PM (stack and fugitive)	China
Zhang et al.	Particulate emission risks of HEVs during cold-starts and warm-restarts: particulate number under fresh and aged aftertreatment systems	<i>J Hazard Mater</i>	Vehicle emission experiment	Particle number; particle number size distribution; sub-23 nm PN	China
Other clinical endpoints					
Yang et al.	Associations of long-term exposure to multiple air pollutants with chronic pain among middle-aged and older adults in China	<i>J Pain</i>	Cohort	PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂ , CO (5-pollutant mixture, WQS)	China

Provenance and method

Entry window **4 August 2026**, contiguous with the 3 August issue. Records were screened on per-record entry date rather than on the harvest window: the sweep ran 4–5 August, but every PubMed record carries an Entrez date of 4 August or earlier and every Crossref deposit a 4 August **created** date, so all 13 belong to this issue and none was held back for the next one. **Sources.** PubMed E-utilities and the PubMed connector on two separate axes (health, instrumentation) windowed on [EDAT] — 10 health and 2 sensing hits. Europe PMC **CREATION_DATE** — 2 hits, 1 in scope. Crossref scoped by journal ISSN across eight aerosol and atmospheric titles that PubMed does not index — 6 hits, all new to the corpus, 2 carried. Consensus on four sensing axes (calibration and drift, satellite–surface fusion, network siting, mobile sensing) — 3 new records after DOI resolution and dedup; the free tier caps returns at 3 per sweep, which remains the binding constraint on the sensing leg. OpenAlex remains blocked behind a paid plan; arXiv returned nothing in scope after the recency screen. **Screening.** Deduplication keys PMID → lowercased DOI → title signature, checked before summarising. **Five records were excluded as off-topic** (agricultural nitrogen accounting, tobacco harm-perception surveys, rice trace gas fluxes, cough-triggered ventilation, surface-water ecological risk) and **five were held, not rejected**: their DOIs verify in Crossref, but no abstract is retrievable from Crossref, Europe PMC or Consensus and the Elsevier and IEEE landing pages are client-rendered. Held records are deliberately absent from **seen.json** so a later issue can carry them once indexed. Nothing was summarised from a title alone, and no DOI was inferred from a journal prefix. **Labels.** Subtopic, design, particle-metric and geography assignments are single-label judgements from title, abstract, keywords and MeSH terms, not a controlled vocabulary. Effect estimates are transcribed verbatim; see the forest-plot caption for why both are co-pollutant rather than PM estimates. All DOI links resolve to the publisher of record and were checked by the automated DOI gate before this issue compiled. Abstracts remain the copyright of their publishers.